

Resection Surgery for Partial Epilepsy. Relation of Surgical Outcome with Some Aspects of the Epileptogenic Process and Surgical Approach

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Summary

In spite of the progressive improvement of the results of resective surgery for epilepsy, the number of not significantly benefited patients remains high. An attempt was made to find out a relation between outcome and some aspects of the pathophysiological organization of the epileptogenic process and of the surgical procedure. Chi-square and logistic regression statistic analyses were utilized. The study was retrospectively performed on 138 surgically treated patients having a minimum follow-up of three years. Three classes of surgical outcome were considered: completely seizure free (including aura; 86 cases, 62.3%), significant seizure reduction (31 cases, 22.5%), and no significant improvement (21 cases, 15.2%). What follows was brought into evidence by the study. 1) On the diagnostic side, the spatial arrangement (focal, unilateral, multifocal) of both the interictal and the ictal epileptic electrocerebral activities are significantly associated with the surgical outcome. Their relative impact on outcome is related to the presence of a structural lesion: when a lesion is documented, the interictal activity has the higher value; vice versa, when no lesion is apparent, the role of the ictal activity is prevalent. However, the presence, as well as the nature of the lesion, per se, are not significantly associated with outcome. 2) On the surgical side, the extent of resection of both the structural lesion and of the epileptogenic zone are highly associated with the surgical result; the extent of lesion resection prevails on that of the epileptogenic zone. The type of surgical approach (hemispherectomy: 17 cases; temporal lobectomy: 67 cases; extratemporal resection: 54 cases) has no significant relation to the outcome. The value and the limits of the results obtained are discussed.

Keywords: Partial epilepsy; resective surgery; prognostic factors; statistics.

Introduction

Outcome of surgical treatment for epilepsy has improved in recent years. Such an improvement is likely to be related to several factors, such as better knowledge of the basic epileptogenic mechanisms, the development of clinical and instrumental diagnostic means, and the refinement of the surgical technique. Never-

theless, the percentage of patients who do not significantly benefit from surgery still remains relatively high. With the exception of hemispherectomies in children, this percentage varies from about 10 to 40 per cent of the treated cases, with large variations in the different series (see for recent reviews: ^{24, 46, 62}). These variations might reflect differences in the local policy of patient selection. Restrictive selection criteria certainly increase the probability of satisfying results. Incompleteness of the diagnostic work-up or misinterpretation of the diagnostic findings are likely to be responsible for some of the surgical failures. Attempts have been made to find which factors could predict the surgical result. Age, sex and occupational status; neurological, psychological and psychiatric status; aetiology; structural and metabolic cerebral abnormalities; duration of the disease; frequency and clinical pattern of the seizures; distribution of the epileptiform activity in scalp and invasive EEGs; location of the epileptogenic process. The results obtained have been recently reviewed^{18, 46, 62}. Discrepancies are quite apparent. Furthermore, seemingly convincing data tend to lose their significance when submitted to statistical analysis¹⁸.

That which has been stated above prompted us to review our personal series. Taking into account what has been reported in the literature and the result of a preliminary personal study, we concentrate on finding a correlation between surgical outcome and some aspects of the pathophysiological organization of the epileptogenic process and of the surgical procedure.

Material and Methods

One hundred and thirty-eight epileptic patients submitted to resective surgical treatment in our Institute were considered. Their age ranged

from 7 months to 46 years (mean 21.6 years); 100 were males and 38 were females. They had frequent and uncontrolled seizures. The duration of epilepsy was of at least three years standing, with the exception of 6 children, 5 of whom underwent hemispherectomy. Seizure type included: complex partial ($n = 98$), simple partial ($n = 37$) and secondarily generalized seizures ($n = 75$). The presence of a causative cerebral structural lesion was documented in 103 cases. The nature of the lesion is reported in Table I. Malignant tumours and AVMs were excluded. Surgery was performed in all patients according to epileptological criteria, aiming at removing both the causative cerebral pathology and the cerebral zone giving origin to the seizure⁵⁰. Lesionectomy alone and callosotomy were not considered.

What follows was included in the diagnostic work-up. 1) Family and personal history, neurological and psychological examination. 2) CT scan and/or (since 1980) MRI; angiography in selected cases. 3) Long-term scalp EEG monitoring (10–20 International System of electrode placement), in most of the “temporal” patients integrated with zygomatic or sphenoidal or foramen ovale electrodes, aiming at detecting both interictal and ictal activity; the EEG recording was accompanied by video monitoring for seizure analysis (at least 3 seizures were recorded in 98 patients). 4) Invasive EEG studies: they were i) acute (electrocorticography 74 cases, stereo-EEG 4 cases), and/or ii) chronic (subdural grids 5 cases, stereo-EEG 43 cases), the latter being always accompanied by TV monitoring (the technique and the criteria developed by Talairach *et al.*⁶¹, were used for the stereo-EEG). 5) Intracarotid Amytal test (23 patients). In several cases the interictal epileptic signals recorded from the scalp or directly from the brain were submitted to automatic elaboration with the main purpose of facilitating the study of quantitative and time related regional differences^{44,48–50}. Particular attention was paid to the search for state independent spiking (wakefulness and sleep:^{49, 52–54}).

The surgical procedures utilized were: 17 hemispherectomies (children only), 67 anterior temporal lobectomies; and 54 extratemporal resections. Hemispherectomy was performed in one block except for the hemimegalencephalic patients, in whom the abnormal hemisphere was removed in three blocks. It was extracaudate in all cases; an external drainage from the subdural cavity¹⁵ was maintained for 4–7 days. Temporal lobectomy was performed in one block in most of the patients; the anterior mesial temporal structures were always removed. In the extratemporal epilepsies the resection volume was defined by taking into account the location of the causative structural lesions (present in 49 out of 54 patients), the information provided by the clinical seizure pattern analysis and the presurgical and/or intrasurgical direct brain recording. When necessary, the functional properties of the affected and nearby cortex were checked with pre- or intra-operative functional cortical mapping. The majority of patients were operated on under general anaesthesia. Local anaesthesia was used in adult patients only and mainly when functional mapping was necessary (16% of the adult subjects).

The evaluation of the surgical result was related to the first three post-surgical years. The problem of surgical outcome's stability in the subsequent years^{22, 24, 45, 57} was not faced. Partial seizures occurring in the first two postoperative weeks were not considered. The surgical outcome was subdivided into three classes. Class I: complete disappearance of seizures, including aura. We decided not to include in this class the patients who still had auras though relieved from overt seizures, because in our opinion the persistence of the aura indicates that the epileptogenic neuronal population was not completely removed at surgery. Class II: auras only or rare seizures (less than 3 partial seizures or absences per year) and “significant”

psychosocial adjustment, as related to family, school or work environment. Class III: all the other cases. The seizure occurrence as well as the psycho-social adjustment was checked at intervals of six months to one year. Full pharmacological medication was maintained for the first two post-surgical years. Thereafter it was gradually tailed in the Class I patients. If seizures recurred it was resumed. In most of these cases low dosages were sufficient to maintain the patient seizure free.

What follows was considered for a correlation with the surgical outcome.

1) *Factors related to the diagnostic work-up.* i) Recognition of the causative pathology, as resulting from pre-surgical neuroradiology. ii) Nature of such a pathology, as revealed by the histological study of the surgical specimen. iii) Spatial distribution of the interictal epileptiform electrocerebral signals (spikes, sharp-waves and spike-and-wave discharges), as resulting from scalp, sphenoidal, and direct cerebral recording. Three main spatial patterns were recognized: a) “monofocal”: at least 80% of interictal epileptiform abnormalities confined to a given cerebral region; b) “unilateral” (*idem*, but with uncertain regional location); c) “multifocal” or “bilateral” (less than 80% focalization or lateralization); no subdivision was made between synchronous and asynchronous bilateral epileptiform abnormalities. iv) Site of origin of the ictal electrocerebral activity associated with the occurrence of the clinical seizure (change of the background activity and appearance of rhythmic spiking or of slow waves of variable amplitude and frequency, or of low-voltage fast activity⁵⁸). Three main patterns were considered: a) “monofocal”, indicating the origin of all the electrophysiologically recorded ictal activity from the same and circumscribed cerebral site; b) “regional”, indicating that all the electrical ictal discharges originated from the same cerebral region which, however, was relatively widespread and not identifiable with a single structure or functional zone; c) “multifocal”, indicating that ictal discharges were recorded at least once from different sites of the same hemisphere or from the two hemispheres. The modality of spread of the ictal discharges was not considered in the present study. All the electrophysiological material (see iii and iv above) was at first screened by one observer (G.C.), and subsequently submitted to a collegial analysis, involving at least three people.

2) *Factors related to the surgical treatment.* i) Type of surgical approach: hemispherectomy, temporal lobectomy and extratemporal resection. ii) Extent of the causative cerebral structural lesion removal, as evaluated by integrating the written report of the surgeon and the post-surgical CT or MRI, which was performed at least one month after surgery. iii) Extent of resection of the epileptogenic zone: it was obviously based on the previous evaluation of the location and size of the epileptogenic zone. The information provided by the simultaneous recording of the electrocerebral ictal activity and of the clinical seizure pattern was regarded as the most reliable means of achieving this purpose in 98 patients. In these cases, the spatial arrangement of the EEG interictal abnormalities was utilized as an integrative type of information. It became of paramount importance in the 40 patients lacking sufficient documentation on the electrocerebral ictal activity. The role of the interictal epileptiform abnormalities was strengthened by automatic analysis as well as by the relation of their topography with the clinical seizure pattern and with the structural cerebral lesion. It is to be noticed that the lesion was lacking in only 2 of the 40 patients, both of them treated by temporal lobectomy. Postsurgical electrocorticography was not available in enough patients to allow statistical evaluation. The extension of the resection of the structural lesion and of the epilep-

Table 1. *Nature of Cerebral Structural Lesion. 103 patients*

Scar-Gliosis		Tumour		Malformation	
Trauma	15	low-grade oligod.	13	hemimegalenceph.	7
Infection	12	low-grade astrocyt.	8	cort. dysplasia	5
Ischaemia	7	hamartoma	2	microangioma	5
Haemorrhage	3	ganglioglioma	1	Sturge-Weber	3
Not defined	22				
59		24		20	

Table 2. *Type of Surgery vs. Outcome (min. follow-up: 3 years)*

	I			II		III	
H	17	13	76.5%	4	23.5%	—	—
TL	67	44	65.7%	14	20.9%	9	13.4%
ER	54	29	53.7%	13	24.1%	12	22.2%
	138	86	62.3%	31	22.5%	21	15.2%

H hemispherectomy, *TL* temporal lobectomy, *ER* extratemporal resection. $p = 0.207$.

Table 3. *Type of Lesion vs. Outcome*

	I			II		III	
Scar	59	34	57.6%	13	22.1%	12	20.3%
Tumour	24	20	83.3%	4	16.7%	—	—
Malform.	20	13	65.0%	6	30.0%	1	5.0%
	103	67	65.1%	23	22.3%	13	12.6%

$p = 0.055$.

togenic zone was classified as “extensive”, “incomplete” and “uncertain”.

The consistency of the association of the factors listed above in 1 and 2 with the surgical outcome was statistically analyzed. All the factors were compared with the surgical outcome using the chi-square statistic. A p value equal or inferior to 0.05 was considered statistically significant. Variables reaching statistical significance at the univariate analysis were subsequently submitted to a multivariate procedure using the logistic regression model¹⁴.

Results

1. Surgical Outcome (Table 2)

Eighty-six patients (62.3%) were completely relieved from seizures (class I); in 31 the seizures were significantly reduced (class II: 22.5%); in the remaining 21 patients the outcome was regarded as not satisfactory (class III: 15.2%). On the whole, 117 patients (84.8%) benefited from surgery. The permanent surgical complications were: 7 mild contralateral motor deficits and

9 visual field deficits. Transitory complications occurred in 20 patients: 11 scalp infections, 6 hemiparesis, 3 hydrocephali requiring a CSF shunt in hemispherectomized children.

2. Correlation Between Surgical Outcome and Diagnostic Factors

i) *Presence of a causative cerebral structural lesion*: as mentioned above, it was detected in 103 patients (74.6%: see Table 1). The percentage of the best results was higher in these patients (65.1% of class I) than in those without evidence of a structural lesion (35 cases: 54.3% in class I). Conversely, the percentage of surgical failures (class III) was higher in the latter. These relations, however, were not statistically significant. ii) *Nature of the cerebral structural lesion* (Table 3). The highest percentage of the best results was obtained in

Table 4. *Interictal Epileptic Activity vs. Outcome*

	I			II		III	
Focal	69	55	79.7%	10	14.5%	4	5.8%
Unilateral	29	13	44.8%	11	37.9%	5	17.25
Multif. Bil.	40	18	45.0%	10	25.0%	12	30.0%
	138	86	62.3%	31	22.5%	21	15.2%

p = 0.000.

Table 5. *Ictal Epileptic Activity vs. Outcome*

	I			II		III	
Focal	67	46	68.7%	13	19.4%	8	11.9%
Regional	18	8	44.4%	8	44.4%	2	11.1%
Multif. bil.	13	3	23.1%	2	15.4%	8	61.5%
	98	57	58.1%	23	23.5%	18	18.4%

p = 0.000.

Table 6. *Interictal and Ictal Epileptic Activities vs. Outcome*

Interictal	Ictal			Outcome
	Focal	Multif.	Regional	
Focal	34	—	2	I
Multif.	10	2	—	
Unilat.	2	1	6	
Focal	7	—	—	II
Multif.	3	2	3	
Unilat.	3	—	5	
Focal	1	1	—	III
Multif.	5	7	—	
Unilat.	2	—	2	

the tumoural cases (83.3% in class I). The scar-gliosis group recognized the highest incidence of unsatisfactory outcome (20.3% in class III). The association between nature of the lesion and surgical outcome was at the limits of the statistical significance ($p = 0.055$).

iii) *Interictal epileptiform electrocerebral activity* (Table 4). A "monofocal" pattern was found in 50% of the patients (69 out of 138). The hemispherectomized children showing a lateralization of at least 80% were included in this group rather than in the "unilateral" one, because of the peculiar pathophysiological organization of the epileptic process and related surgical

approach; the 5 remaining hemispherectomized patients belonged to the "bilateral" group (i.e., less than 80% lateralization). The "monofocal" group was the one with the highest percentage of class I surgical outcome (79.7%). The highest incidence of poor results (class III) was found in the "bilateral or multifocal" group (30.0%). The association was highly significant ($p = 0.000$). It must be noticed, however, that in the group of patients with "bilateral" or "multifocal" interictal epileptic abnormalities a satisfactory surgical outcome (class I plus class II) was prevalent also (70%).

iv) *Ictal epileptiform electrocerebral activity* (Table 5). It could be evaluated in 98 patients. Fifty-two of them underwent pre-operative invasive EEG studies. A structural lesion was identified in 65 cases. A "monofocal" location of the onset of repeatedly recorded electrocerebral ictal discharges leading to clinical seizures was found in 67 patients, a "regional" location in 18, and a "multifocal" or "bilateral" one in 13. Excellent results (class I) were obtained in 68.7% of the patients in the "monofocal" group, in 44.4% in the "unilateral" group and in 23.1% in the "multifocal" one. The latter was affected by the highest rate of surgical failure (61.5%). These differences were statistically highly significant ($p = 0.000$).

v) *Convergence of the interictal and ictal epileptiform electrocerebral activities* (Table 6). It was evaluated in 98 patients (see iv above). The topographical arrangement of the interictal and of the ictal activities were congruent in 66

Table 7. *Extent of Lesion Removal vs. Outcome*

	I			II		III	
Complete	63	54	85.7%	8	12.7%	1	1.6%
Incomplete	31	8	25.8%	12	38.7%	11	35.5%
Uncertain	9	5	55.6%	3	33.3%	1	11.1%
	103	67	65.1%	23	22.3%	13	12.6%

p = 0.000.

Table 8. *Extent of Epileptogenic Zone Resection vs. Outcome*

	I			II		III	
Extensive	71	57	80.3%	11	15.5%	33	9.2%
Incomplete	33	15	45.5%	8	24.2%	10	30.3%
Uncertain	34	14	41.2%	12	35.3%	8	23.5%
	138	86	62.3%	31	22.5%	21	51.2%

p = 0.000.

cases (67.3%). The “monofocal” distribution of the interictal activity, which was found in 45 patients, was congruent with the ictal focality in 42 cases (93.3%); 34 of them (79.5%) had an excellent surgical outcome, 7 were of class II and only 1 of class III. On the other hand, out of 18 patients with a focal ictal pattern, but a multifocal interictal distribution, 10 had a class I, 3 a class II and 5 a class III outcome. The relation between congruence of the two types of epileptiform electrocerebral events and surgical outcome was highly significant ($p = 0.000$). vi) *Convergence of the electrocerebral and morphological findings.* It was analyzed in the 59 patients having in common the presence of a structural lesion and the recording of both the interictal and ictal epileptiform electrocerebral abnormalities. In 38 patients (58.5%) there was a good convergence of the topographical arrangement of both types of electrocerebral epileptiform abnormalities (“monofocal” and “regional” or “unilateral”) and of the location of structural lesion. Complete seizure relief was achieved in 73.7% of cases and significant improvement in 18.4%, no significant improvement was limited to 7.9% of patients. In 14 patients (21.5%) the location of the structural lesion was concordant with that of only the interictal (the ictal being “multifocal”) or of the ictal (the interictal being “multifocal”) activity. Eleven of them (78.6%) still benefited from surgery (7 class I: 50.0%; 4 class II: 28.6%), while about one fifth were not significantly improved (class III: 21.4%). Finally,

in the remaining 7 patients (10.8%) both the types of electrocerebral epileptiform abnormalities were “multifocal”, while the structural lesion was spatially delimited; however, one of the “foci” was found close to the lesion and it was resected together with the lesion. Two entered the outcome class I; 2 the class II and 3 the class III. In no cases did the electrophysiological and morphological findings conflict. (e.g., electrophysiological abnormalities confined or prevalent in the hemisphere contralateral to the one harbouring the lesion, or in distant sites of the same hemisphere).

3. Correlation Between Surgical Outcome and Factors Related to the Surgical Treatment

i) *Type of surgical approach* (Table 2). The best results were obtained in the hemispherectomized patients (76.5% seizure free: class I); the less satisfying results followed the extratemporal resections (53.7% in class I, and 22.2% in class III). However, the type of surgical procedure and outcome did not turn out to be significantly associated. ii) *Extent of the cerebral structural lesion removal* (103 patients: Table 7). It was regarded as “complete” in 63, “incomplete” in 31, and not sufficiently defined in 9 of the 103 patients with a known structural lesion. The surgical outcome was very good in the complete resection group (85.7%). Conversely, only one fourth of the patients (25.8%) became seizure free when the removal was incomplete. The association

was highly significant ($p = 0.000$). iii) *Extent of the epileptogenic zone resection* (138 patients: Table 8). It was regarded as “extensive” in 71 patients (51.4%), and “incomplete” in 33 (23.9%). All the 13 “multifocal” or “bilateral” patients of 2, iv above (see Table 5) obviously belong to the “incomplete” group; they were submitted to the resection of only one of the multiple epileptogenic zones; in the remaining patients of the group, the incompleteness of resection was related to the involvement of functionally relevant cortical areas (as for instance left temporal cases with the epileptogenic zone extending posteriorly, or extratemporal cases with involvement of the motor frontal speech areas). In 34 (24.6%) patients the surgeon was not sure of the completeness of the removal (the “uncertain” group); that happened when the extension of the epileptogenic zone could not be sufficiently defined. As was to be expected, the best surgical results (class I) were obtained in the “extensive” resection group (80.3%), while the highest incidence of disappointing surgical outcome (class III) was found in the “incomplete” resection group (30.3%). It is to be noted, however, that 45.5% of the “incomplete” resection group became seizure free. The association between surgical outcome and extent of resection of the epileptogenic zone was highly significant ($p = 0.000$). iv) *Correspondence of the extent of resection of the structural lesion and epileptogenic zone*. It was found in 59 patients. Forty-two of them underwent a “complete” or “extensive” resection of both the structural lesion and of the epileptogenic zone: 39 (92.9%) had a class I outcome; the other 3 patients (7.1%) fell into class II; no one in class III. In 17 cases the resection was “incomplete” both as regards the lesion and the epileptogenic zone: 7 (41.2%) had a class I outcome, 5 (29.4%) class II and 5 class III. The association with the outcome was highly significant ($p = 0.000$).

4. Relative Impact on Surgical Outcome of the Factors Considered

The diagnostic and surgical variables which significantly affected the surgical outcome at the chi-square analysis (see 2 and 3 above) entered the multivariate procedure. A binary outcome was considered to utilize the stepwise logistic regression analysis. The surgical results were subdivided in two classes only: one including the cases with complete seizure suppression (the class I of the previous analysis), and the other grouping patients who improved and who did not significantly benefit from surgery (the classes II and III of previous

analysis). Furthermore, the study was limited to the patients showing all the variables to be analyzed.

i) *Diagnostic variables*: the two variables considered were the spatial arrangement of the electrocerebral epileptiform interictal and ictal activities, both of which associated very significantly with the outcome at the chi-square analysis. Both variables were available in 98 patients. The stepwise logistic regression analysis showed an obvious prevalence of the interictal epileptiform activity ($p = 0.0011$) over the ictal one, which actually lost its significance ($p = 0.2437$). The patients with (65 cases) and without (33 cases) a structural lesion were then separately considered. In the patients with a lesion, the prevalence of the interictal epileptic activity was maintained ($p = 0.008$ versus $p = 0.071$ of the ictal one); vice versa, in the patients without evidence of a structural lesion, the impact on outcome of the ictal activity became highly significant ($p = 0.001$), while that of the interictal activity was lost ($p = 0.074$). ii) *Surgical variables*: the extent of resection of the structural lesion and that of the epileptogenic zone were significantly associated with the surgical outcome. Both of them retained their high significance when matched one against the other (103 patients), though with a slight prevalence of the first ($p = 0.000$) over the second ($p = 0.001$). iii) *All significant variables*: spatial location of the interictal and of the ictal epileptiform activities, and extent of removal of the structural lesion and of the epileptogenic zone were all submitted to the stepwise regression analysis. That could be done in 65 patients only; at the chi-square analysis all of the four variables maintained their significance. The multivariate analysis showed a definite prevalence of the extent of the lesion removal ($p = 0.000$), followed by the spatial arrangement of the interictal epileptiform activity ($p = 0.013$); the spatial arrangement of the ictal epileptic discharges and the extent of resection of the epileptogenic zone lost their significance ($p = 0.086$ and $p = 0.2157$, respectively).

Discussion

The success of any type of treatment, be it surgical or not, for any type of disease is obviously related to several factors. As far as the surgical treatment of the epilepsies is concerned, the factors considered were many, as mentioned in the introduction. Some of them, however, may not be recognized or are difficult to analyze, as for instance the so-called “predisposition” to epilepsy, namely the relative facility to develop epilepsy that the brain of certain patients seems to manifest

once injured, independently of the nature of the injury. The 7 factors considered in the present study were all characterized by their direct relationship to the pathophysiological process at the time of surgery and to surgery itself.

The presence of a structural lesion, the nature of the lesion and the type of surgical approach did not appear to be significantly associated with the surgical outcome when submitted to the chi-square statistical analysis. Nonetheless, the incidence of the better results was higher: i) in patients harbouring a structural lesion, ii) when the nature of the lesion was neoplastic, and iii) in the children treated by hemispherectomy. A highly significant association with the surgical outcome was shown by two factors belonging to the diagnostic work-up and by two, related to the surgery: i) focal distribution of the interictal epileptic electrocerebral activity and ii) of the site of origin of the ictal activity; iii) completeness of the structural lesion removal and iv) extent of the epileptogenic zone resection. The convergence of both the diagnostic and the surgical factors significantly increased the likelihood of the best outcome.

The attempts we made to find out the relative impact on outcome of the significant factors listed above indicates that: i) the most relevant of the diagnostic variables is the degree of focalization of the epileptiform electrocerebral activity recorded between seizures; ii) as for the surgical variables, the extent of resection of the structural lesion and of the epileptogenic zone have a similar but not an equal impact, the first being slightly more significant; iii) when all four significant variables, diagnostic and surgical, are simultaneously analyzed, the extent of resection of the structural lesion reveals to have the greatest impact on outcome. Some of the results could be expected; others less.

1) *The prevalence of the interictal over the ictal electrocerebral activity* in predicting the surgical outcome might give rise to perplexities: the diagnostic value of the interictal epileptic activity is certainly relevant^{4, 8, 11, 13, 16, 17, 32, 34, 35, 38, 41, 52, 55}. According to some⁶⁵, when utilizing long-lasting direct brain recording, the interictal epileptiform activity provides a more useful definition of the epileptogenic area than the ictal activity. However, the findings by Talairach and co-workers in the early seventieths⁶¹ led one to regard the detection of the site of origin of the epileptic discharges leading to the clinical seizure as the most reliable electrophysiological channel of information. Indeed, false lateralizing interictal spiking has been repeatedly reported^{2, 3, 7, 15, 26, 38, 63, 69}. We believe that this view is correct⁵⁰,

and we acted accordingly with the majority of our patients (71%). However, we also believe that there are cases in whom the combination of the analysis of the interictal epileptic activity with other diagnostic findings can be sufficient for a correct planning and carrying out of surgery⁵¹; see also ⁴ and ¹³). The diagnostic role of the interictal epileptiform activity has to be searched for with appropriate analysis^{49, 52-54}. It is increased by the accurate quantitative evaluation and by the study of the persistence (or stability) in time, with relative independence of the physiological state of the patient (e.g., wakefulness and the different phases of sleep) and of the pharmacological medication, as well as by the time relation between spikes recorded from different sites. The value of the interictal epileptic activity is supported by the significant decrease in the probability of a good outcome when a monofocal ictal pattern is accompanied by a multifocal interictal one (see p. 105). It is in cases like these, and particularly if lacking a structural lesion, that the feasibility of surgery and the planning of the surgical approach depend mainly on the information provided by the study of the ictal activity and its correlation with the clinical seizure pattern. It is reasonable to assume that these are the most "difficult" cases.

To try to give value to this hypothesis, we looked for the incidence of "incomplete" resection of the epileptogenic zone in the 98 patients undergoing study of the ictal activity, and we compared it with that of the remaining 40 patients operated upon on the indication given by the interictal epileptiform activity (obviously correlated with the clinical and neuroradiological information). The incidence of "incomplete" epileptogenic zone resection was definitely higher in the first group (28 patients: 28.6%) than in the second (5 patients: 12.6%); the difference was statistically significant (0.045). The incidence of poor results was also higher, as would be expected. As mentioned above, the recognized incompleteness of the epileptogenic zone resection was related in most of the cases to the partial involvement of eloquent cortical areas. The relatively poorer surgical outcome in the first group of patients, therefore, does not necessarily imply a poor localizing value of the electrocerebral ictal activity. It rather indicates that several of these patients required all the possible types of diagnostic information because of the necessity of a carefully tailored surgical resection. Further support for the above formulated hypothesis comes from the different association with outcome of the two electrocerebral variables considered in the patients with a structural lesion and in those lacking it (see p. 106):

the role of the ictal activity prevails over that of the interictal one in the patients without a lesion, and vice versa.

2) *The high relevance of the causative structural cerebral lesion removal* in addition to that of the epileptogenic cerebral tissue has been stressed by many^{1, 10, 19, 21, 25, 29, 32, 37, 39, 40, 42, 59, 66}, including ourselves⁵⁰, though not by all^{47, 65}. Our study, in addition, indicates on a statistical basis that even if the presence of a structural lesion is not significantly associated with the surgical outcome, when present the impact of its removal on the surgical outcome is superior to that of the resection of the epileptogenic zone. A similar conclusion was reached by Palmini *et al.*⁴² in 26 cases of cortical dysplasia and by Van Ness⁶² in extratemporal epilepsies. This finding seems to strengthen the role of the "lesionectomy" in the surgical treatment of epilepsy. Several studies have shown that, indeed, surgery limited to the removal of the lesion without intentional resection of the epileptogenic tissue can bring seizure relief^{12, 20, 27, 29, 30, 33, 57}. Our personal experience in about 80 lesionectomy cases confirms such a possibility (in preparation). One must remark that, so far, in most of the patients treated with "lesionectomy", including ours, the lesion was of neoplastic origin. According to the present study, and to those of others^{35, 47}, the tumoural epilepsies have a better outcome than those related to other lesions. Our present analysis, however, indicates that such a good surgical prognosis is related more to the possibility of achieving a complete lesion removal than to the tumoural nature of the lesion. Our findings by no means imply that the surgical resection of the epileptogenic zone is scarcely relevant. First of all, it obviously remains the only valid surgical means when there is no evidence of a causative structural lesion (25.4% of patients of our present series); second, the association of the resection of the epileptogenic zone with the outcome is highly significant (see Table 8); third, the best outcome is obtained if the resection of the epileptogenic zone is combined with that of the structural lesion (92.9% in our series; see also^{5, 16, 33, 43, 65}); fourth, resection limited to the epileptogenic zone with sparing of the structural lesion can bring about good results^{27, 28, 60}. The real problem is that the extent of resection of the epileptogenic zone is not easy to define. It depends on the knowledge of the spatial configuration of a functional, not an anatomical entity, the "boundaries" of which cannot be rigid, but fluctuate. In spite of the diagnostic effort, doubts may remain²³. That seems confirmed by our relatively high number of "uncertainties" on the extension of resection

(24.6% of cases, see Table 8). To be noted as well is that the surgical failure in 9.20% of our patients in whom the epileptogenic zone was regarded as "extensively" removed (Table 8) is likely to be related to an erroneous evaluation of its spatial arrangement. On the other hand, the complete disappearance of seizures in 80% of them indicates that our criteria of evaluation were correct in most of the cases. The disappearance of seizures in 45% of the patients classified as having an "incomplete" epileptogenic zone resection is also interesting. It may emphasize the likelihood of an erroneous evaluation; however, it might also indicate that the reduction of the neuronal population giving origin to the epileptic discharges responsible for the seizure can be sufficient to cure the patient. Other means of study, such as the P.E.T. and the S.P.E.C.T., seem apt to provide an additional line of information integrating with the electrophysiological one (see for recent reviews: ⁶ and ³¹). We have no experience with the P.E.T., and that with the S.P.E.C.T. is not yet sufficient for an evaluation.

We believe that the removal of the structural cerebral lesion and that of the epileptogenic zone are both of paramount importance for the success of the surgical treatment. Consequently, the accurate presurgical search for the spatial location of both the components of the epileptogenic process remains the most important objective of the presurgical diagnostic work-up; the finding of a convergence of the electrophysiological and morphological studies has the highest predictive value^{9, 10, 36}. While the reasons of the high therapeutic role of the epileptogenic zone resection can be easily understood, that of the lesion removal can only be ventured: i) at least part of the epileptogenic neurons might be intermingled with the non-neuronal pathological tissue (be it neoplastic, or reparative, or dysmorphic; ii) closely located epileptogenic cerebral tissue might be included in the resection of the lesion; iii) the lesion might produce a sustained but reversible enhancing action on the epileptogenicity of the neurons, perhaps due to a breakdown of the blood brain barrier⁵⁶. These hypotheses might also apply to the frequently advocated "lesionectomy" as an efficient surgical means to treat epilepsy. The point is that, up until now at least, there is no way to foresee whether the above hypothesized events will occur or not, or will be sufficient to eradicate the epilepsy.

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